

SEDAR 47 Southeastern U.S. Goliath Grouper

Assessment Team

The Florida Fish and Wildlife Conservation Commission held a workshop on March 14-16, 2016 to discuss recent research findings about Goliath Grouper in Florida waters. Before the close of the workshop, the participants provided their recommendations about additional research that should be conducted on this species to improve our understanding of this species.

Monitoring activities

- Genetics: sample from fish from around Florida, and particularly the Florida West Coast. Samples could be from removal of a few scales, fin clips, or needle biopsy. Consider training at-sea observers/samplers to collect these samples. Eggs could also be collected and analyzed. A repeat of the recent kinship analysis (Tringali) on a periodic basis (5-10 years) would help monitor for changes in the degree of relatedness in the Florida Keys and southeast Florida.
- Spawning aggregations – locate additional sites where aggregations occur, using a combination of sound and Didson sonar imaging to verify spawning activity. This is work currently in progress. Monitor currently known spawning sites for trends over time.
- Mark-recapture data needs to be analyzed from the acoustic tagging data and about 800 sampled and visually tagged fish on the east and west coast of Florida. Investigate the possibility of using genetic mark-recapture methods.
- Expand sampling for nursery habitat and targeted juvenile sampling, possibly using an existing fishery-independent sampling program. Recommend to the NMFS Cooperative Research Program the possible funding of projects to work with the blue crab trap fishermen to collect fin clips (for genetics) when there is bycatch of Goliaths.
- Annual age sampling on the level of 400-500 specimens to monitor age structure of adults. The fin ray-age validation work is in progress.
- Fecundity research – in progress.
- Investigate the use of wildlife models like occupancy modelling. This may require more regular, systematic sampling than is currently available.
- Use visual data from the REEF survey, NMFS-UM Reef Visual Census (though they do not sample artificial reefs and wrecks), and expand the Great Goliath Grouper Counts from once a year in June to twice a year (June and September) to help identify locations with larger fish to sample.
- Drop cam video from FWRI's FIM program could expand the coverage of visual surveys, but would need to expand sampling to artificial reefs/wrecks.
- Investigate feasibility of mounting video cameras on charter and head boats to obtain information on bycatch (some preliminary work by Mote Marine Lab may be useful).
- Discuss with the FWC Artificial Reef Program the possibility of grant funding for Goliath work.

- Promote the collection of Goliath lengths from anglers (Snook and Game Fish Foundation app)
- Use GIS artificial reef data to identify all artificial reef structures and related data (materials, heights) in the Gulf of Mexico for developing a sampling plan.
- Extract dates and locations from log book data especially during spawning season that may identify new aggregations/spawning sites.

Review Workshop

The data used in the models mostly originated from Florida. Sparse data from elsewhere in the species (historical) range may be indicative of low population size (either as a function of natural distribution patterns or constriction of the population due to heavy fishing pressure), or poor sampling (including landings). This issue needs to be further explored as it has bearing on the geographical validity and usefulness of the assessment for regional management.

There was some concern by the RP about the method the Assessment Team used for combining the GGGC and the REEF survey as there is a potential for bias (e.g. potential for targeting sites with known high abundance of Goliath Grouper in the GGGC survey). How influential the inclusion of the GGGC data was to the outcome of the model should be explored.

Many of the research recommendations provided in the Assessment Report include research that would not necessarily improve future assessments for this species. The SEDAR 23 RW concluded that “The next benchmark assessment cannot be successfully completed without data from the research recommended by the Data, Assessment, and Review Panels.” The outcome of the SEDAR 47 benchmark assessment process indicates that much of this information is still needed in order to successfully complete an assessment for Goliath Grouper.

Specifically, research and monitoring efforts that could improve future assessments for Goliath Grouper include:

Life history information

Basic reproductive data is lacking throughout the species distribution. This includes size and age at maturity for each sex, sexual sequence with size and age for each sex, and fecundity. In the SEDAR 47 assessment, the reproduction functions used in the models made some strong assumptions about the maturity schedule and fecundity rates that were based on insufficient data. Greater resolution of data, especially maturity at size or age, would alleviate the impact of these assumptions for future assessments.

A limited research harvest should be considered to fill the remaining gaps in life history information for Goliath Grouper. Such a harvest should incorporate individuals from across the size spectrum, but should focus on larger individuals as they may be beneficial to ground truth the fin-ray aging techniques used for the offshore age composition, and to develop fecundity schedules.

Additional research on the age structure of the catch, especially in the offshore recreational fishery, is needed. The SEDAR 47 assessment used age composition of only 22 adult individuals that were caught by a research fishery and aged with fin rays (Koenig et al. 2013). This age

composition was used for multiple parts of the assessment and may provide a large source of the assessment uncertainty. Cooperative research efforts with the recreational charter and headboat fisheries could be informative towards generating better information on the offshore recreational age composition.

Discard mortality estimates are needed across the species distribution. For the SEDAR 47 assessment, a fixed discard mortality estimate was applied to the post-moratorium harvest. However, the uncertainty around this estimate is unknown and may be substantial.

Stock definition

SEDAR 23 recommended that Goliath Grouper should be genetically sampled from areas across the stock range in the South Atlantic and Gulf of Mexico to allow for a more thorough examination of the current single stock definition. The SEDAR 47 RW was presented with a brief summary of these efforts, which seem to support that single stock definition. Like many other sources of information informing the SEDAR 47 assessment, this information remains in progress or is incomplete and has not yet been vetted by peer review.

Examination of spawning aggregations over the entire distribution range should include seasonality, sex ratios, and individual fidelity.

Fishery independent sources of information are lacking or uncertain

The SEDAR 47 AT indicated that a specifically designed pre-fishery recruit survey (e.g. mangrove habitat) would help guide recruitment in the assessment model.

Develop and/or explore methods to take into account episodic mortality events.

One issue with the SEDAR 47 assessment was the use of a fixed value for natural mortality at age, despite evidence that episodic mortality events (i.e. cold-kills) have affected the Goliath Grouper population. Options to account for this mortality should be explored for future assessments. Methods used in other assessments (e.g. to address red tide events affecting red and gag grouper in the GOM) include incorporating episodic mortality events as a separate removal fleet. These methods may be appropriate for Goliath Grouper and could reduce some of the uncertainty in the estimates of natural mortality.

Reexamine methods of constructing historical removals

The use of length data from MRFSS/MRIP recreational Goliath Grouper removals need to be further examined. In SEDAR 47, the methods used to apply mean length of catch was inconsistent between years when there was missing and/or suspect data, and years with an estimate from the MRFSS/MRIP database. This introduced a significant amount of uncertainty to the harvest estimates.

Incorporate Data from Low Abundance Years into Indices

The Assessment Team discarded some of the data from index development due to very low catch rates in years adjacent to the moratorium. As a result, low abundance indices are removed from the assessment. Methods for incorporating these data into appropriate statistical models for standardization and development of indices should be explored.